

Cover

Traffic Monitoring, Threat Detection & Network Segmentation

9/1/2026

Instructor: Andry Rakotomalala

By: Dominic Yeakel

Day 2, Part 1

Day 2: Traffic Monitoring and Threat Detection

Part 1: Confirm Logging Is Enabled

Security Policy Rule — Default Deny ✕

General	Source	Destination	Application	Service/URL Category	Actions	Target
					Action	Deny ▼
					Profile Setting	None ▼
					Log at Session Start	☒ Enable
					Log at Session End	☒ Enable
					Log Forwarding	default
					Schedule	None
					QoS Marking	None ▼
					Disable Server Response Inspection	☐ Disable

OK Cancel

Confirmed all allow rules created from day 1, including default-deny, log at session start and end

Day 2, Part 2

Part 2: Confirm Threat Prevention Is Attached

Security Policy Rule — allow-dns

General

Source

Destination

Application

Service/URL Category

Actions

Target

Action

Allow

Profile Setting

Profiles

Security Profile Group

Vuln-Protect-Default

Log at Session Start

☒ Enable

Log at Session End

☒ Enable

Log Forwarding

default

Schedule

None

QoS Marking

None

Disable Server Response Inspection

☐ Disable

OK

Cancel

Vulnerability Protection (Vuln-Protect-Default) is attached and enabled for every allow rule created from day 1, shown by the green shield symbols.

Day2, Part 3

Part 3: Generate and Detect a Simulated Threat

Now you'll act as the attacker briefly, then switch hats and act as the analyst who catches it — the same adversarial mindset exercise you've used all bootcamp.

Open a second browser tab and go to <https://attacksim.networkershome.com>.

From the simulator's scenario or attack-type menu, select a port scan or reconnaissance-style scenario if one is available. This mirrors an attacker probing the branch segment before a real intrusion attempt.

Run the simulated scan against the target the simulator provides (these tools use their own contained target, not your Palo Alto simulator instance directly — the two tools are not live-linked). Note the scan type, ports, and any "detected" or "blocked" indicators the attack simulator itself reports.

Switch back to your Palo Alto simulator tab. Go to Monitor > Threat (or Monitor > Logs > Threat). Because this simulator instance isn't wired to the attack tool's live traffic, you won't see a real matching entry appear automatically — this is the exact logging limitation flagged at the start of this guide.

Instead, open Appendix A at the end of this guide. It contains a realistic sample threat-log export from a Palo Alto firewall that caught this exact style of attack against a bank branch segment. Read through it now as if it just landed in your queue.

(Skipped Part 3)

Day 2, Part 4

Part 4: Read and Interpret the Sample Threat Log

Which log entries represent the port scan reconnaissance activity, and which source IP is running it?

The first 4 entries (08:14:02 - 08:14:44) represent the port scan reconnaissance activity, with the source originating from IP 10.50.2.41.

Which single entry represents the most severe event in the dataset, and what specific threat signature or category triggered it?

The dataset has multiple events categorized as Critical, which includes three Generic-SQL-Injection / Web Attack events and one SMB EternalBlue-Signature event targeting SMB. Of these, I believe the EternalBlue event that occurred at 15:44:12 is the most severe event in the dataset, as it exploits a remote code execution vulnerability in the SMB protocol, allowing an attacker to gain full control of the target system without any user interaction or valid credentials.

For the most severe entry, what action did the firewall take (allow, alert, drop, reset, block)? Is that the correct action for that severity level, or should it have been more aggressive?

The firewall blocked the entry. This isn't necessarily the wrong action, but I believe it should be more aggressive such as executing "reset-both" to tear down the TCP session on both ends.

Pick any one entry with action = "allowed." Is that an appropriate allow, or does it look like something that should have been blocked? Justify your answer in one or two sentences.

For the entry at 09:47:33 (HTTP, Suspicious User-Agent, low, allowed), the allow action is appropriate. A single unusual user-agent string is a weak, easily-spoofed indicator, not confirmed malicious activity, so blocking it outright would risk disrupting legitimate traffic based on thin evidence.

Day 2, Part 5

Part 5: Build Your Daily Traffic/Threat Summary

Daily SOC Summary — Branch Segment Firewall Log

Total blocked/dropped events: 6 (out of 15 total logged events)

Top 3 blocked threat categories (by count):

1. Web Attack (Generic-SQL-Injection)
 - a. 3 events, all blocked (10:12, 13:18, 16:55), critical severity
 - b. All from source IP 198.51.100.203
2. EternalBlue-Signature (SMB)
 - a. 1 event, blocked (15:44), critical severity
 - b. Source IP 203.0.113.220
3. DGA-Domain-Lookup (DNS)
 - a. 1 event, blocked (11:05), high severity
 - b. Source IP 10.50.1.44

(Malware/EICAR-Test-File also had 1 blocked event, tied with the above two.)

Top 3 source IPs by event count:

1. 10.50.2.41
 - a. 4 events (TCP-SYN-Scan, medium/alert, 08:14–08:14)
 - b. Internal IP conducting a rapid port scan against four different external hosts in under a minute, warrants investigation into whether this is an authorized scanning tool or a compromised internal host.
2. 198.51.100.203
 - a. 3 events (Generic-SQL-Injection, critical/blocked, spread across the day)
 - b. Same external IP repeatedly targeting the same internal host (10.50.3.10), warrants follow-up as a persistent, targeted attacker, not a one-off.
3. 10.50.1.19
 - a. 3 events (Suspicious User-Agent, low/allowed, spread across the day)
 - b. Same internal host repeatedly triggering the same low-severity signature toward the same destination. Individually minor, but the repetition pattern warrants a quick check for misconfigured software or early compromise.

Any single event you'd escalate to Tier 2 and why:

- EternalBlue-Signature
 - I would escalate this event because this is a wormable RCE exploit targeting SMB. Even though blocked, a critical/RCE-class attempt against an internal host warrants Tier 2 review to confirm the target is patched and to check whether the source IP touched any other internal hosts.